L5: Divichlet's Theorem and applications of the FTA

Theorem 1.21 (Dirichlet's Theorem on primes in AP): Let $a, b \in \mathbb{Z}$ with $a, b > 0$ and $(a, b) = 1$. Then the arithmetic progression (AP)
$a, a+b, a+2b, \ldots, a+n,$
contains infinitely many primes.

- Unfortunately, the proof of Dirichlet's Theorem is outside the scope of this class.
  Remark: Setting $a = b = I$ recovers the result we proved about the infinitude of prime numbers.
- We can use the FTA to prove special cases of Dirichlet's Theorem e.g. $a = 3, b = 4$.
  Proposition 1.22: There are infinitely many primes of the form $4n + 3$ where n is a non-negative integer.
- First, we need the following Lemma.

Lemma).23: Let $a, b \in \mathbb{Z}$. If a and b are expressible in the form $4n+1$, where n is an integer, then ab is also expressible in that form.
Proof: Let $a = 4n_1 + 1$ and $b = 4n_2 + 1$ with $n_1, n_2 \in \mathbb{Z}$. Then

$$\begin{aligned} ab &= (4n_1 + 1)(4n_2 + 1) \\ &= 16n_1n_2 + 4n_1 + 4n_2 + 1 \\ &= 4(4n_1n_2 + n_1 + n_2) + 1 \\ &= 4n + 1, \end{aligned}$$

where $n = 4n_1n_2 + n_1 + n_2 \in \mathbb{Z}$.
Proof (of Prop.1.22): Assume, by way of contradiction, that there are finitely many prime numbers of the form $4n + 3$ where n is a non-negative integer, say

$$P_0 := 3, P_1, P_2, \cdots, P_r.$$

Here, the prime numbers $P_1, \cdots, P_r$ are assumed to be distinct from $P_0 = 3$. Consider the
integer $N = 4P_1 \cdots P_r + 3$. The prime factorisation of N must contain a prime number p of the desired form or it would otherwise contain only prime numbers of the form $4n+1$ where n is a positive integer, and hence be of the form $4n+1$ where n is a positive integer by Lemmal.23.
Lo $p = p_0 = 3$ or $p = p_i$ for some $i = 1, 2, \ldots, r$.
Case I $(p = p_0 = 3)$: Then 31 N implies that $3 \mid N - 3$ by Proposition 1.2, and we have that $314P_1P_2 \cdots P_r$. By Corollary 1.15, we now have that 314 or 31 Pi for some $i = 1, 2, \ldots, r$. Contradiction I

Case II ($p = p_i$ for some $i = 1, 2, \ldots, r$): Then $P_i \mid N$ implies that $P_i \mid N - 4P_1P_2 \cdots P_r$ by Proposition 1.2, and we have that $p_i \mid 3$, a contradiction! Thus there are infinitely
many prime numbers of the desired form.

Remark: The proof of Proposition 1.22can$'$ 't be extended to prove Dirichlet's Theorem in full generality i.e. Lemma 1.23 does not hold for integers of the form $4n+3$ (exercise for home).

Congruences

- Up to the beginning of the 19$^{\text{th}}$ century number theory consisted of brilliant results but with no connecting theme.
- In 1801, Gauss in "Disquisitiones Arithmeticae" unified number theory $\rightarrow$ and developed the theory of congruences. Gauss was only 24 years of age! Definition: Let $a, b, m \in \mathbb{Z}$ with $m>0$. Then a is said to be congrnent to b modulo m, denoted $a \equiv b(\bmod m)$, if $m \mid a-b$. If $a \equiv b(\bmod m)$, then m is said to be the modulus of the congruence. The notation $a \neq b(\bmod m)$ means a is not congment to b modulo m i.e. $m + a-b$.
  Remark: The modulus in a congruence is defined to be a positive integer. dometimes this is assumed implicitly.
  Example: $\cdot 4 \mid 25-1$, so $25 \equiv 1(\bmod 4)$
- I la-b for all $a, b \in \mathbb{Z}$, so $a \equiv b(\bmod \pm)$ for all $a, b \in \mathbb{Z}$.
- $5 \not\mid -7-2$, so $-7 \not\equiv 2(\bmod 5)$

Proposition 2.1: Congruence modulo m is an equivalence relation on $\mathbb{Z}$. Proof: Let $a \in \mathbb{Z}$. Since any integer divides 0 , we have m / O, from which $m / a-a$ and $a \equiv a(\bmod m)$. Thus Congruence modulo m is reflexive on $\mathbb{Z}$. Let $a, b \in \mathbb{Z}$ and assume that $a \equiv b(\bmod m)$. Then $m \mid a-b$, and (6) so we have $m \mid(-1)(a-b)$, or equivalently, $m(b-a$. Thus $b \equiv a(\bmod m)$, and congruence modulo m is symmetric. Let $a, b, c \in \mathbb{Z}$ and assume that $a \equiv b(\bmod \mathrm{m})$ and $b \equiv c(\bmod m)$. Then $m \mid a-b$ and $m \mid b-c$, and so $m /(a-b)+(b-c)$ by Proposition 1.2. Thus $m \mid a-c$, and so $a \equiv c(\bmod m)$. Thus congruence modulo m is transitive on $\mathbb{Z}$. We have proved that congruence modulom is an equivalence velation on $\mathbb{Z}$.

Consequence 2.2: $\mathbb{Z}$ is partitioned into equivalence classes under congruence modulo m.

- For a given $x \in \mathbb{Z}$, let $[x]$ denote the equivalence class of x modulo m. This should not be confused with the floor function.

Example: The equivalence classes of $\mathbb{Z}$

under congruence modulo 4 are

$$
\begin{aligned}
[0] &= \{x \in \mathbb{Z} : x \equiv 0(\bmod 4)\} \\
&= \{x \in \mathbb{Z} : 4 \mid x - 0\} \\
&= \{x \in \mathbb{Z} : 4 \mid x\} \\
&= \{x : x = 4n \text{ for some } n \in \mathbb{Z}\} \\
&= \{\ldots, -8, -4, 0, 4, 8, \ldots\},
\end{aligned}
$$

and similarly,
$[1] = \{x : x = 4n + 1 \text{ for some } n \in \mathbb{Z}\}$

$$= \{\ldots, -7, -3, 1, 5, 9, \ldots\}$$

$[2] = \{x : x = 4n + 2 \text{ for some } n \in \mathbb{Z}\}$.

$$= \{\ldots, -6, -2, 2, 6, 10, \ldots\}$$

$[3] = \{x : x = 4n + 3 \text{ for some } n \in \mathbb{Z}\}$

$$= \{\ldots, -5, -1, 3, 7, 11, \ldots\}$$

$\mathbb{Z}$ is partitioned into four equivalence classes
under congruence modulo 4 .

- From this example we see that any integer is congruent to $0, 1, 2$, or 3 modulo 4 .
  Definition: A set of integers such that every integer is congrnent modulo m to exactly one integer of the set is said to be a Complete residue system modulo m.
  Example: - $\{0, 1, 2, 3\}$ is a complete residue system modulo 4.

- $\{6, -11, 19, 1988\}$ is also a complete residue Syrtem modulo 4.
  i.e. $6 \equiv 2(\bmod 4), -11 \equiv 1(\bmod 4), 19 \equiv 3(\bmod 4)$, $1988 \equiv 0(\bmod 4)$.
  Proposition 2.3 : The set $\{0, 1, 2, \ldots, m-1\}$ is a complete residue system modulo m.
  Proof: We first show that every integer is congruent modulo m to at least one
  integer in $\{0, 1, 2, \ldots, m-1\}$. Let $a \in \mathbb{Z}$. (9) Then by the Dividion algorithm, there exist $q, r \in \mathbb{Z}$ such that

$$a = mq + r, \quad 0 \leq r < m.$$

Now $a = mq + r$ implies that $a - r = mq$, so $m \mid a - r$, or equivalently, $a \equiv r(\bmod m)$.
Furthermore, $0 \leq r < m$ implies that $r \in \{0, 1, 2, \ldots, m-1\}$, which shows the initial claim.

We now show that every integer is congruent modulo m to at most one integer in $\{0, 1, 2, \cdots, m-1\}$, which would complete the proof. Let $a \in \mathbb{Z}$ and

$r_1, r_2 \in \{0, 1, \ldots, m-1\}$ such that $a \equiv r_1(\mathrm{mod} m)$ and $a \equiv r_2(\mathrm{mod} m)$.
Then $r_1 \equiv r_2(\mathrm{mod} m)$, or equivalently $m \mid r_1 - r_2$. Since $r_1, r_2 \in \{0, 1, \ldots, m-1\}$, we have that

$$-(m-1) \leq r_1 - r_2 \leq m-1.$$

Since $m \mid r_1 - r_2$, we have that $r_1 - r_2 = 0$ (10) i.e. $r_1 = r_2$. Thus every integer is congruent modulo m to at most one integer in
$\{0, 1, 2, \ldots, m-1\}$.
Definition: The set $\{0, 1, 2, \ldots, m-1\}$ is said to be the set of least non-negative residues modulo m.

Proposition 2.4 : Let $a, b, c, d \in \mathbb{Z}$ such that $a \equiv b(\mathrm{mod} m)$ and $c \equiv d(\mathrm{mod}\ m)$. Then
(a) $a + c \equiv b + d(\mathrm{modm})$
(b) $ac \equiv bd(\mathrm{mod} m)$

Proof: Since $a \equiv b(\mathrm{mod} m)$ and $c \equiv d(\mathrm{mod} m)$, we have that $m \mid a - b$ and $m \mid c - d$. For (a), note that $m \mid (a-b) + (c-d)$, which is equivalent to $m \mid (a+c) - (b+d)$, from which we have

$$a + c \equiv b + d(\mathrm{mod} m).$$

For (b), note that $m/a - b$ implies that
$m \mid c(a-b)$, and that $m \mid c - d$ implies that (11) $m \mid b(c-d)$. Thus $m \mid c(a-b) + b(c-d)$, which is equivalent to $m \mid ac - bd$, and so

$$ac \equiv bd(\mathrm{mod} m).$$